

MOHAN DEVENDRA

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EDUCATION

GITAM UNIVERSITY

Master of Science in Data Science

Visakhapatnam, India

Jul 2024 - May 2026

HINDUSTAN SHIPYARD DEGREE COLLEGE, ANDHRA UNIVERSITY

Bachelor of Science (Mathematics, Statistics & Computer Science)

Visakhapatnam, India

Jul 2020 - Jul 2023

SKILLS

Programming & Data Analysis: Python, SQL, TypeScript, Java, NumPy, Pandas, scikit-learn

Machine Learning & AI: TensorFlow, Keras, Transfer Learning (VGG19-UNet, VGG16, ResNet50), Classification, SVM, KNN, Grad-CAM, GridSearchCV

Web & Backend: Next.js, FastAPI, Flask, PostgreSQL, SQLite, Streamlit, Docker, Selenium, BeautifulSoup

Data Visualization: Matplotlib, Seaborn, Tableau, Plotly Dash, Folium, Recharts, IBM Cognos

Cloud & Tools: AWS (SageMaker, S3), Azure, Supabase, Vercel, Railway, Hugging Face Spaces, Salesforce

Languages: English (Fluent), Telugu (Native), Hindi (Conversational)

INTERNSHIPS

IBM SKILLSBUILD INTERNSHIP (AI/ML DOMAIN)

Data Science Intern

Visakhapatnam, India

Jun 2025 – Aug 2025

- **ML Pipeline Development:** Designed and implemented a complete supervised learning pipeline data cleaning, feature engineering, model training, and evaluation achieving **86.7% accuracy** with Gradient Boosting, outperforming baseline Logistic Regression by 1.4 percentage points.
- **Model Comparison & Selection:** Systematically benchmarked Logistic Regression, Random Forest, and Gradient Boosting using **Target Encoding** and hyperparameter tuning; selected GBM based on F1-score and cross-validation stability.
- **Production Deployment:** Shipped a live inference app on **Streamlit Cloud** using a feature-compressed model (5 inputs, 81.6% accuracy) that reduced prediction latency for real-time user requests.

SALESFORCE VIRTUAL INTERNSHIP

Salesforce Developer Intern

Visakhapatnam, India

May 2025 – Sep 2025

- Engineered automated CRM workflows and custom dashboards using **Apex** and **Lightning Web Components**, enhancing data visibility for business operations.
- Designed process automation solutions that reduced manual data entry time, validated through comprehensive testing on the Trailhead platform.

AWS AI-ML VIRTUAL INTERNSHIP (AWS ACADEMY & EDUSKILLS)

Machine Learning Intern

Visakhapatnam, India

Sep 2022 – Nov 2022

- Architected an end-to-end ML pipeline using **AWS SageMaker** and **S3**, handling data ingestion, preprocessing, and model deployment in a cloud environment.
- Implemented scalable cloud-based training workflows, optimizing resource usage for large datasets during a rigorous 10-week technical program.

UNIVERSITY PROJECTS

CINEPHILE TV – FULL-STACK TV ANALYTICS PLATFORM

- **Production Architecture:** Built a self-initiated TV analytics platform ingesting **4.6M IMDb data points** with a separated **Next.js 14 frontend (Vercel)** and **FastAPI backend (Railway)** backed by PostgreSQL (Supabase); 64 production deployments live.
- **Database & Scraping Engine:** Designed custom **PostgreSQL SQL functions** for top-250 ranking, full-text search, and genre-based recommendations; built a **Selenium scraper fallback** for live episode ratings on shows outside the database.
- **Features & UX:** Delivered interactive **episode rating heatmaps**, Top 250 rankings, a Hall of Fame covering 406,782 episodes, and a **TypeScript + Tailwind CSS** responsive UI optimised for mobile and desktop.

M.SC. THESIS – GREEN SPACE INEQUITY ANALYSIS IN MUMBAI USING DEEP LEARNING

- Trained a custom **VGG19-UNet semantic segmentation model** on 8,910 Pleiades-1A satellite image patches (0.5 m/px) to classify 6 urban land cover classes, achieving **93.72% accuracy** and 94.79% weighted F1, outperforming DeepLab v3+ and SegNet benchmarks.
- **Key Finding:** Quantified a **fourfold green space disparity** informal settlements (Dharavi) average **9.02% GOS coverage** vs **37.95%** in planned residential areas providing the first satellite-derived measurement of green space inequity in Mumbai.
- **Full-Stack Deployment:** Deployed model via **Flask REST API on Hugging Face Spaces (Docker)** with a **Next.js frontend (Vercel)** featuring a tile-stitch pipeline for any-resolution GeoTIFF input and real-time GOS index computation.
- **Training Methodology:** Used two-phase transfer learning (ImageNet encoder freeze then fine-tune), **Focal Loss ($\gamma=2.0$)** for class imbalance, and custom class weights ($2.0\times$ for informal settlements) trained on Google Colab T4 GPU.

FALCON 9 FIRST STAGE LANDING PREDICTION (IBM APPLIED DATA SCIENCE CAPSTONE)

- Engineered an end-to-end machine learning pipeline to predict rocket landing success, processing live data from **SpaceX APIs** and **Web Scraping** to estimate launch cost reductions.
- Developed an interactive **Plotly Dash** web application and **Folium** geospatial maps, enabling real-time visualization of launch success rates across different global sites.
- Benchmarked four classification models (SVM, KNN, Logistic Regression, Decision Tree), achieving **88.93% accuracy** with the top-performing Decision Tree model tuned via **GridSearchCV**.
- Conducted extensive Exploratory Data Analysis (EDA) using **SQL** and Pandas to identify critical correlations between payload mass, orbit type, and mission outcome.

CERTIFICATIONS

Google Advanced Data Analytics – Google	Jun 2025
IBM Data Analyst – IBM	Mar 2025
Meta Data Analyst – Meta	Jul 2025
Microsoft Azure Data Scientist Associate (DP-100) – Microsoft	Aug 2025
Google IT Automation with Python – Google	Jan 2025
Advanced Python: Working with Data – LinkedIn Learning	Feb 2025